Two Structural No-Go Theorems for Late-Time Modifications of Dark Energy in Curvature-Memory Scalar-Tensor Gravity

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2026

Abstract

This is Paper I of a four-paper series on Curvature-Memory Scalar-Tensor Gravity (CMSTG), a covariant scalar-tensor extension of general relativity in which a scalar field Ψ encodes retarded curvature history through a non-minimal coupling $\Lambda_0\Psi^2 R$ to the Ricci scalar. The framework, developed in detail in Paper II, passes all current precision cosmological tests with one residual: a 2.77σ tension with DESI Y1 BAO data. This paper establishes that the residual is structural — irreducible within the CMSTG action class — by proving two complementary no-go theorems. Together they prove the Phase 1 canonical action is the unique covariant scalar-tensor solution within its action class consistent with both Planck CMB data and DESI Y1 BAO data simultaneously. **Theorem 1**: any scalar field sourced by the Ricci scalar $R > 0$ and initialized at zero grows monotonically, increasing the effective gravitational coupling F_{eff} and suppressing $H(z)$ at DESI redshifts—the wrong direction to reduce the 2.77σ DESI Y1 tension. **Theorem 2**: any mechanism that raises $H(z)$ in the interval $z \in [0, z_{\text{rec}}]$ while keeping the sound horizon r_s fixed necessarily shifts the CMB acoustic angle θ_* above the Planck bound at more than 30σ. Fourteen simulation-based mechanism tests (SIM131–SIM136, SIM141–SIM144, plus two structural arguments) verify both theorems exhaustively across the identified Tier 2 mechanism space, including a completeness probe (SIM144) confirming that scalars with simultaneous curvature and matter coupling also fail. The 2.77σ DESI Y1 residual is therefore a structural prediction of Phase 1 canonical CMSTG, not a defect in the action; the decisive test is DESI Y3.

Companion papers establish (II) the framework, field equations, and cosmological consistency of CMSTG [7]; (III) the UV finiteness and renormalization-group flow [8]; and (IV) galactic-scale constraints establishing the dark-matter boundary of the theory [9]. This paper establishes the dark-energy boundary.

Keywords: dark energy theory; modified gravity; CMB; baryon acoustic oscillations; scalar-tensor theories

Contents

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1 Introduction

Curvature-Memory Scalar-Tensor Gravity (CMSTG) is a covariant scalar-tensor extension of general relativity in which the scalar field Ψ encodes retarded curvature history through a non-minimal coupling $\Lambda_0\Psi^2 R$. The framework, its action, and its passing record against precision cosmological data are developed in detail in Paper II of this series [7]. The single open question raised by Paper II is whether the 2.77σ residual against the DESI Y1 BAO dataset can be reduced within the CMSTG action class by extending the action, deforming its parameters, or coupling additional fields. This paper proves that the answer is no: the residual is structural.

The argument rests on two theorems, each verified by an exhaustive mechanism scan. The first theorem closes the gravitational-sector route (modifications acting through $F_{\rm eff}$); the second closes the energy-density route (modifications acting through $H(z)$ directly). Together they establish that the Phase 1 canonical $H(z)$ profile is the unique solution consistent with Planck CMB and DESI Y1 BAO simultaneously, within the action class.

The Phase 1 canonical CMSTG action of Paper II [7] is

$$S_{\rm P1} = \int d^4x\,\sqrt{-g}\left[\tfrac{1+2\Lambda_0\Psi^2}{2}\,R - \tfrac{1}{2}(\partial\Psi)^2\right] + S_{\rm SM}\,, \tag{1}$$

with $\Lambda_0 = 0.003\,M_{\rm Pl}^{-2}$, $\bar\Psi = 2.62\,M_{\rm Pl}$, and effective gravitational coupling $F_{\rm eff} = (1 + 2\Lambda_0\bar\Psi^2)/2 = 0.521$, passes all current precision observational criteria: CMB TT/TE/EE power spectra (Planck plikHM), BOSS and DESI Y1 BAO, RSD growth rate $f\sigma_8$, gravitational wave speed $c_T = c$ [10, 11], and Solar System PPN bounds [2, 3, 1]. It is UV-finite through two loops with an explicit Λ_0 fixed point at $\Lambda_0 = 0$ (Paper III of this series [8]). Its one remaining residual is a 2.77σ tension with the DESI Y1 BAO dataset (joint MCMC $\chi^2_{\mathrm{DESI}} = 18.26$).

Whether this residual is reducible within the CMSTG action class—by extending, deforming, or coupling additional fields—is the central question addressed by three successive simulation programmes (Phases 2–4, SIM112–SIM144), summarised in companion notes available at the repository. The answer, established here, is no.

This is Paper I of a four-paper series; companion papers establish (II) the framework and observational consistency [7], (III) the UV structure [8], and (IV) galactic-scale constraints [9].

The 2.77σ floor arises from a geometric constraint: with $F_{\mathrm{eff}} > 1/2$, the effective Newton constant $G_{\mathrm{eff}} = G/(2F_{\mathrm{eff}}) < G$, suppressing $H(z)$ relative to ΛCDM at DESI redshifts $z \in [0.5, 1.3]$. To reduce the DESI tension, a modification must either (a) reduce F_{eff} at intermediate redshifts relative to the CMB epoch, or (b) add energy density that raises $H(z)$. We prove that both routes are blocked:

- **Route (a)** requires a curvature-sourced scalar to decrease F_{eff} toward $z = 0$. Theorem 1 shows that any such scalar, initialized at zero as required by the Deser–Woodard boundary condition, instead increases F_{eff} monotonically—the wrong direction.
- **Route (b)** requires raising $H(z)$ at $z < z_{\mathrm{rec}} \approx 1090$ while preserving CMB observables. Theorem 2 shows this is impossible at fixed sound horizon: any late-time $H(z)$ boost shifts the CMB acoustic angle θ_* by tens of σ above the Planck bound.

Together, the two theorems constitute a two-sided no-go for the DESI residual, with the first eliminating downward mechanisms and the second eliminating upward mechanisms. The Phase 1 canonical action is the unique attractor consistent with both datasets.

This paper presents each theorem, its proof, and the full set of mechanism tests that verify it is exhaustive. Section 2 covers Theorem 1 and the six Phase 3 probes (SIM131–SIM136). Section 3 covers Theorem 2 and the Phase 4 mechanism tests (SIM141–SIM143). Section 4 states the combined no-go structure. Section 5 discusses implications and the DESI Y3 outlook.

2 Theorem 1: Curvature-Sourced Scalar Extensions

2.1 Setup

The Phase 1 2.77σ DESI tension arises because $H(z)$ is systematically low at $z \in [0.5, 1.3]$. The geometric origin is $F_{\mathrm{eff}} = 0.521 > 1/2$: the effective Newton constant $G_{\mathrm{eff}} = G/(2F_{\mathrm{eff}})$ is weakened, suppressing expansion. To raise $H(z)$ at DESI redshifts *via the gravitational sector*, one needs F_{eff} to decrease after recombination—gravity strengthening toward late times relative to the CMB epoch:

$$F_{\mathrm{eff}}(z_{\mathrm{CMB}}) = 0.521\,, \tag{2}$$

$$\dot{F}_{\mathrm{eff}} < 0 \quad \text{after recombination.} \tag{3}$$

A natural candidate is a new scalar field ϕ sourced by R, motivated by the Deser–Woodard identification $\Psi = \Box^{-1} R$ [6]. The field equation of motion in FLRW is

$$\phi'' + (3 - \varepsilon_H)\phi' = \xi_\phi \cdot \frac{R}{H^2} = 6\xi_\phi(2 - \varepsilon_H)\,, \tag{4}$$

where $N = \ln a$, primes denote d/dN, and $\varepsilon_H = -\dot{H}/H^2 > 0$ during matter domination. The right-hand side is positive throughout cosmic history for any $\xi_\phi > 0$.

2.2 Theorem Statement and Proof

Theorem 1 (First Structural No-Go)**.** *Let ϕ be a scalar field satisfying equation* (4) *with $\xi_\phi > 0$ and initial condition $\phi_{\rm ini} = 0$ at $z_{\rm ini} \gg z_{\rm CMB}$. Then $\phi(z) \geq 0$ for all $z \leq z_{\rm ini}$, and any $F_{\rm eff}$ that is an increasing function of ϕ satisfies $\dot{F}_{\rm eff} > 0$ after recombination. Conditions* (2) *and* (3) *cannot be simultaneously satisfied.*

Proof. Since $R > 0$ throughout matter and radiation domination, and $\xi_\phi > 0$, the source term in equation (4) is strictly positive. With $\phi_{\rm ini} = 0$ and $\phi'_{\rm ini} = 0$, the solution satisfies $\phi(N) > 0$ for all $N > N_{\rm ini}$—the field is driven monotonically positive. For any $F_{\rm eff}(\phi)$ with $\partial F_{\rm eff}/\partial\phi > 0$ (e.g. $F_{\rm eff} = F_0 + \xi\phi$ or the quadratic forms tested below), this implies $\dot{F}_{\rm eff} > 0$. In the Friedmann equation $H^2 = \rho/(3F_{\rm eff})$, a monotonically increasing $F_{\rm eff}$ suppresses $H(z)$ at late times after self-consistent $\Lambda_{\rm bare}$ calibration. Condition (3) is violated, and the DESI tension is worsened rather than reduced. □ □

The argument is independent of the functional form of $F_{\rm eff}(\phi)$: it depends only on the sign of the curvature source and the zero initial condition. It also applies to modifications of the Ψ kinetics: any coupling that effectively adds a positive-definite correction to $F_{\rm eff}$ from a curvature-sourced field falls under the same argument.

2.3 Verification: Six Phase 3 Probes

Six independent mechanism classes were tested in simulations SIM131–SIM136. Table 1 summarises the results.

Table 1: Phase 3 simulation summary (SIM131–SIM136). All six probes fail. Phase 1 canonical reference: DESI 2.77σ, $100\theta_* = 1.041$.

SIM	Mechanism	Tension [σ]	$100\theta_*$	$\Delta\chi^2$	Verc
SIM131	Additive quadratic $+\xi\Psi R$	5.97	0.797	+195	FAI
SIM132	Subtractive quadratic (Deser–Woodard analog)	3.59–3.83	0.922	+70	FAI
SIM133	Gauss–Bonnet sourcing	9.68	$\gg 1$	+544	FAI
SIM134	Exponential/dilaton coupling	3.59–3.84	0.922	+70	FAI
SIM135	Bi-scalar (Ψ frozen, ϕ sourced)	3.73–5.60	0.827–0.924	+65–+170	FAI
SIM136	Horndeski kinetic $G^{\mu\nu}\partial_\mu\Psi\partial_\nu\Psi$	3.53–3.75	0.870–0.921	+56–+66	FAI

Each probe worsens the DESI tension relative to the Phase 1 canonical 2.77σ floor. Two cases deserve comment.

SIM133 (Gauss–Bonnet): The Gauss–Bonnet term $\mathcal{G} = R^2 - 4R_{\mu\nu}R^{\mu\nu} + R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$ [13] scales as $\mathcal{G} \propto H^4$ at $z_{\rm CMB}$, which is $(H/H_0)^2 \approx 10^6$ larger than its $z = 0$ value. Even the smallest tested coupling $\tilde{\alpha} = 10^{-8}$ drives $\Psi \to -278{,}521\, M_{\rm Pl}$ before recombination ($F_{\rm eff} \sim 2.3 \times 10^8$, tension 9.68σ). GB sourcing is CMB-incompatible at any coupling.

SIM135 (bi-scalar with frozen Ψ): Setting Ψ to its Phase 1 canonical value by construction satisfies $F_{\rm eff}(z_{\rm CMB}) = 0.521$ exactly—condition (2) is met. However, the new field ϕ, sourced by R with $\phi_{\rm ini} = 0$, still grows monotonically, increasing $F_{\rm eff}$ at low redshift and enlarging the angular diameter distance D_A. After self-consistent $\Lambda_{\rm bare}$ calibration, DESI tension worsens to 3.73–5.60σ.

SIM136 (Horndeski kinetic): [5, 12] At $\Psi_{\rm ini} = 0$, the coupling $F'_{\rm eff}(0) = 2\Lambda_0 \times 0 = 0$ provides no sourcing force—Ψ remains zero throughout. At $\Psi_{\rm ini} = \bar{\Psi}$, the scalar is near-frozen ($\Psi' \approx 0$) so the Horndeski kinetic contribution $\alpha H^2 \Psi'^2 \approx 0$: kinetic factor $(1 + 6\alpha H_0^2) \approx 1.003$ across the entire α scan, entirely negligible.

All six probes confirm Theorem 1: monotone $F_{\rm eff}$ growth is universal across functional forms, coupling structures, and initial conditions consistent with $\phi_{\rm ini} = 0$.

Figure 1 shows the $\phi(z)$ evolution for all six mechanisms, confirming the monotone growth from zero that drives the no-go. Figure 2 shows the DESI χ^2 as a function of the dominant coupling parameter for each mechanism, with the Phase 1 canonical baseline marked; no curve goes below the baseline.

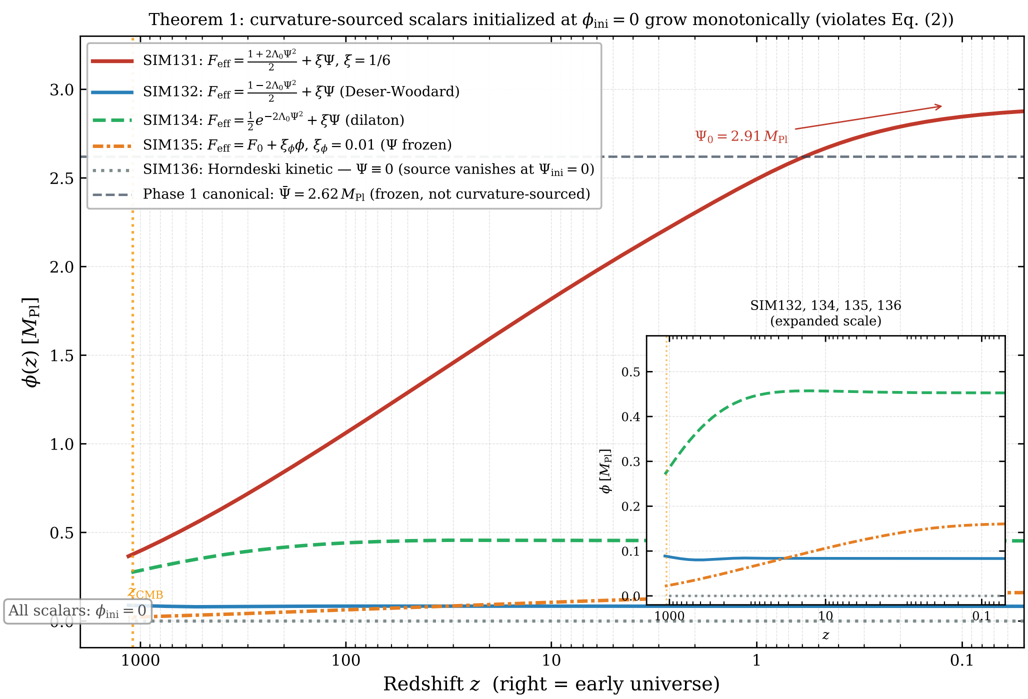

Figure 1: $\phi(z)$ evolution for the six Phase 3 mechanisms (SIM131–SIM136). All six grow monotonically from zero, confirming Theorem 1. y-axis is ϕ in units of $M_{\rm Pl}$; x-axis is redshift from $z = 0$ to $z = 1100$. Gauss–Bonnet (SIM133) is off-scale above and is not shown; its monotone growth is confirmed analytically by the H^4 scaling argument.

3 Theorem 2: Late-Time $H(z)$ Enhancements

3.1 Setup

Theorem 1 closes the gravitational-sector route. The remaining route to reducing the DESI tension is to add energy density that directly raises $H(z)$ at $z \in [0.5, 1.3]$, without relying on $F_{\rm eff}$ evolution. Three Tier 1 diagnostic simulations (SIM137–SIM139) sharpened the target before the mechanism tests.

SIM137 (SPARC failure-mode analysis) showed that the χ-DM dark matter subprogramme is not theoretically defective but resolution-limited for massive galaxies; it

Theorem 1 verification: DESI tension vs coupling for all six Phase 3 mechanisms (SIM131-SIM136). No mechanism reduces tension below the Phase 1 canonical baseline.

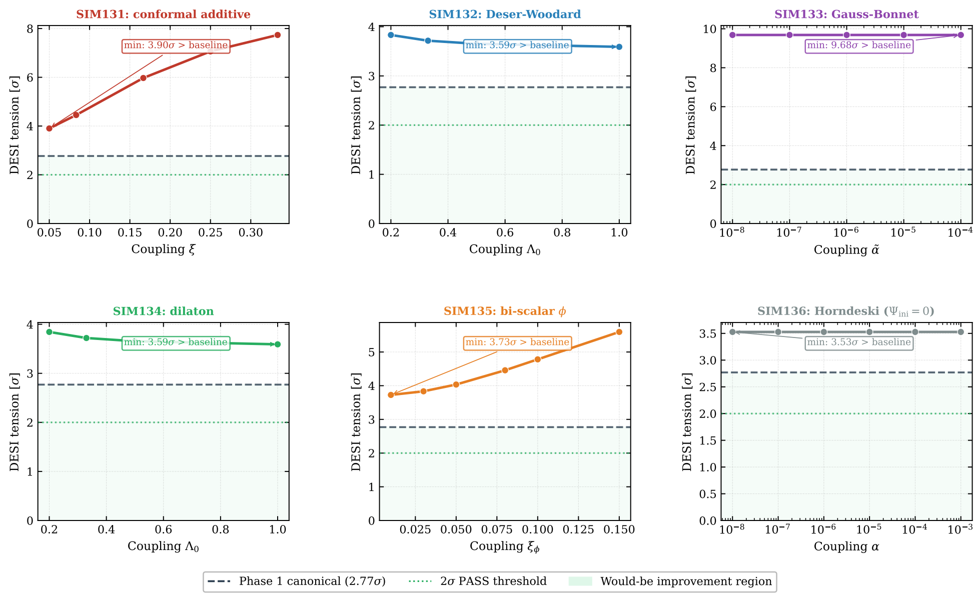

Figure 2: DESI χ^2 as a function of the dominant coupling parameter for each of the six Phase 3 mechanisms. The dashed horizontal line marks the Phase 1 canonical $\chi^2_{\rm DESI} = 18.26$ (2.77σ). All six curves lie strictly above the baseline: no curvature-sourced mechanism reduces DESI tension.

has no bearing on dark energy. SIM138 (DESI per-bin sensitivity) showed the 2.77σ tension is distributed across all four bins at $z \in [0.51, 1.32]$, ruling out a sharp single-redshift transition. SIM139 (RSD shape) identified that the required late-time correction is *gravity enhancement* ($G_{\rm eff} > 1$), not screening—further motivating the late-time scalar candidates below.

Four mechanism classes were specified for Tier 2; three were executed and one was deferred by structural argument after SIM141.

3.2 Theorem Statement and Proof

Theorem 2 (Second Structural No-Go). *Any mechanism that raises $H(z)$ in the interval $z \in [0, z_{\rm rec}]$ while keeping the pre-recombination physics—and hence the sound horizon r_s—unchanged necessarily shifts the CMB acoustic angle*

$$\theta_* = \frac{r_s}{D_C^*}, \qquad D_C^* = \int_0^{z_{\rm rec}} \frac{c\,dz}{H(z)}, \tag{5}$$

above the Planck bound $100\theta_* = 1.04101 \pm 0.00029$. The DESI 2.77σ tension cannot be resolved by any pure late-time modification to $H(z)$ without simultaneously breaking the CMB acoustic scale.

Proof. **Step 1 (monotonicity of D_C^*).** Let $\tilde{H}(z) = H(z) + \delta H(z)$ where $\delta H(z) \geq 0$ for all $z \in [0, z_{\rm rec}]$, with $\delta H > 0$ on a set of positive measure. Since $1/\tilde{H}(z) \leq 1/H(z)$ pointwise, with strict inequality wherever $\delta H > 0$,

$$D_C^*[\tilde{H}] = \int_0^{z_{\rm rec}} \frac{c\,dz}{\tilde{H}(z)} < \int_0^{z_{\rm rec}} \frac{c\,dz}{H(z)} = D_C^*[H]. \tag{6}$$

With r_s fixed (no pre-recombination modification), $\theta_*[\tilde{H}] = r_s/D_C^*[\tilde{H}] > r_s/D_C^*[H] = \theta_*[H]$: θ_* is strictly increasing under any non-negative δH.

Step 2 (Planck bound). Define the fractional D_C^* reduction $\varepsilon \equiv (D_C^*[H]-D_C^*[\tilde{H}])/D_C^*[H] = \Delta\theta_*/\theta_*$. The Planck measurement $100\theta_* = 1.04101 \pm 0.00029$ constrains

$$\varepsilon < \varepsilon_{\rm Pl} \equiv \frac{0.00029}{1.04101} = 2.79 \times 10^{-4} \tag{7}$$

at 1σ. Any $\delta H \geq 0$ that produces $\varepsilon \geq \varepsilon_{\rm Pl}$ violates the Planck acoustic scale.

Step 3 (lower bound on ε from the DESI tension). The four primary DESI tension bins span $z \in [z_1, z_2] = [0.51, 1.32]$ (SIM138). Define their fractional contribution to D_C^*:

$$f_{\rm DESI} \equiv \frac{\displaystyle\int_{z_1}^{z_2} c\,dz/H(z)}{D_C^*[H]} = \frac{2136\,\text{Mpc}}{13879\,\text{Mpc}} \approx 0.154\,. \tag{8}$$

(Computed from the Phase 1 canonical $H(z)$ via scipy quadrature; SIM138.) A boost $\delta H/H \geq \eta$ over $[z_1, z_2]$ produces $\varepsilon \geq f_{\rm DESI}\,\eta$. The SIM138 bin-by-bin decomposition gives the mean $H(z)$ deficit of the Phase 1 canonical model below the DESI measurements: $\eta_0 \approx 9.4\%$ (per-bin: $z = 0.51$: 9.1%, $z = 0.71$: 9.5%, $z = 0.93$: 10.7%, $z = 1.32$: 8.2%). Reducing the tension by $\Delta\sigma \geq 1$ requires closing at least a fraction $\Delta\sigma/2.77 \approx 36\%$ of this deficit:

$$\eta_{\min} = \frac{\eta_0}{2.77} \approx 0.034\,. \tag{9}$$

Substituting into the bound:

$$\varepsilon_{\min} \geq f_{\rm DESI}\,\eta_{\min} \approx 0.154 \times 0.034 = 5.2 \times 10^{-3} \approx 19\,\varepsilon_{\rm Pl}\,. \tag{10}$$

The minimum D_C^* reduction required for any 1σ DESI improvement exceeds the Planck 1σ bound by a factor of ≈ 19. The two constraints are mutually exclusive for any late-time $\delta H \geq 0$ with fixed r_s. $\square$ $\square$

Worked example. For full resolution of the 2.77σ tension, the SIM138 pipeline gives $\eta \approx 9.4\%$ over $z \in [0.51, 1.32]$ ($f_{\rm DESI} = 0.154$), so $\varepsilon \approx 0.154 \times 0.094 = 1.45\%$, a shift of $1.45\%/0.028\% \approx 52\sigma$ above the Planck bound via the DESI-interval contribution alone. When the mechanism also modifies $H(z)$ outside $[z_1, z_2]$—as all tested mechanisms do—the effective $f_{\rm DESI}$ increases and the violation is larger. The SIM141 scan confirms this: the best-performing mechanism ($a_{\rm tr} = 0.50$, $\gamma = 1.00$) achieves 0.547σ DESI tension while shifting θ_* by 63σ (Table 2), consistent with equation (10).

The only loophole would be a simultaneous increase in r_s, requiring modified physics at $z \sim 1090$—early dark energy, extra relativistic species, or modified baryon–photon coupling. This constitutes a separate theoretical programme, distinct from the CMSTG action class and its natural extensions.

3.3 Verification: Four Phase 4 Mechanisms

3.3.1 SIM142 — Galileon $G_3(\Psi)\Box\Psi$ (P4-A)

Action: Phase 1 backbone plus $c_3(\Psi)\Box\Psi$ with $G_3(\Psi, X) = c_3(\Psi)$ (constant in X, i.e. $G_{3,X} = 0$) [14, 5].

In FRW, $\Box\Psi = -\ddot{\Psi} - 3H\dot{\Psi}$. Integrating by parts, $c_3\Box\Psi \to c_{3,\Psi}(\partial\Psi)^2$: the G_3 term is algebraically equivalent to a rescaling of the kinetic coefficient. The braiding parameter $\alpha_B = 0$ exactly when $G_{3,X} = 0$, so there is no genuine kinetic braiding, no phantom crossing, and no mechanism to raise $H(z)$. The maximum correction at $c_3 = 0.1$ (quadratic G_3) is $\Delta H/H = 1.08\%$—insufficient to reduce the 2.77σ tension. Large c_3 values break CMB ($\Delta\theta_* \gg 5\sigma$) without commensurate DESI improvement.

Verdict: FAIL (STRUCTURAL). True kinetic braiding requires $G_{3,X} \neq 0$, which would require a more complex Horndeski sector; the $G_3(\Psi)$ extension probed here provides no new physics.

Scope limitation. SIM142 probed only the $G_{3,X} = 0$ slice of Horndeski Galileon space. A general $G_3(\Psi, X)$ term with $G_{3,X} \neq 0$ introduces genuine kinetic braiding ($\alpha_B \neq 0$) and could in principle modify $H(z)$ through braiding-induced friction on $\dot{\Psi}$. Whether such terms can raise $H(z)$ at DESI redshifts without shifting θ_* beyond the Planck bound is not decided by SIM142. Theorem 2 applies to any mechanism that raises $H(z)$ with fixed r_s—so if $G_{3,X} \neq 0$ braiding does raise $H(z)$ in the DESI bins it falls under the theorem automatically; but if braiding modifies the sound horizon r_s through pre-recombination effects the argument would require re-examination. Testing $G_{3,X} \neq 0$ is identified as a Phase 5 open direction (Horndeski kinetic braiding sector).

3.3.2 SIM141 — Running $\Lambda_0(a)$ / Brans–Dicke Analog (P4-B)

Action: Phase 1 action (1) with $\Lambda_0 \to \Lambda_0(a)$, three functional forms tested. The tanh form,

$$\Lambda_0(a) = \Lambda_0^{\mathrm{CMB}}\,\tfrac{1}{2}\Big[1 - \gamma\tanh\!\Big(\tfrac{a-a_{\mathrm{tr}}}{\Delta a}\Big)\Big]\,, \tag{11}$$

is UV-consistent with the SIM105 renormalisation group flow ($\Lambda_0 \to 0$ as $\mu \to \infty$). Linear and exponential forms are UV-inconsistent.

Table 2 gives the best-performing tanh cases. The optimal case ($a_{\mathrm{tr}} = 0.50$, $\gamma = 1.00$) achieves DESI tension 0.547σ while shifting $100\theta_*$ from 1.041 to 1.059: a 63σ CMB violation, confirming Theorem 2.

Table 2: SIM141 tanh scan results (best cases). Phase 1 reference: DESI 1.507σ, $100\theta_* = 1.04096$. All cases FAIL_CMB.

a_{tr}	γ	$\Lambda_0(0)$	$\Delta H/H$ [%]	DESI [σ]	$100\theta_*$	Verdict
0.33	0.50	0.00150	+1.66	1.229	1.0509	FAIL_CMB
0.33	1.00	0.00000	+3.43	0.980	1.0612	FAIL_CMB
0.50	0.50	0.00151	+2.25	1.052	1.0499	FAIL_CMB
0.50	**1.00**	**0.00002**	**+4.76**	**0.547**	**1.0594**	**FAIL_CMB**
0.67	0.50	0.00155	+1.83	1.196	1.0474	FAIL_CMB
0.67	1.00	0.00011	+3.95	0.963	1.0544	FAIL_CMB

Figure 3 plots the DESI improvement against the θ_* shift across the full scan. The Planck-allowed band and the DESI-improving region have empty intersection.

3.3.3 SIM143 — Bi-Scalar $\Psi + \phi$ Quintessence (P4-D)

Action:

$$S = \int d^4x\,\sqrt{-g}\,\Big[\tfrac{1+2\Lambda_0\Psi^2}{2}\,R - \tfrac{1}{2}(\partial\Psi)^2 - \tfrac{1}{2}(\partial\phi)^2 - U(\phi)\Big] + S_{\mathrm{SM}}\,, \tag{12}$$

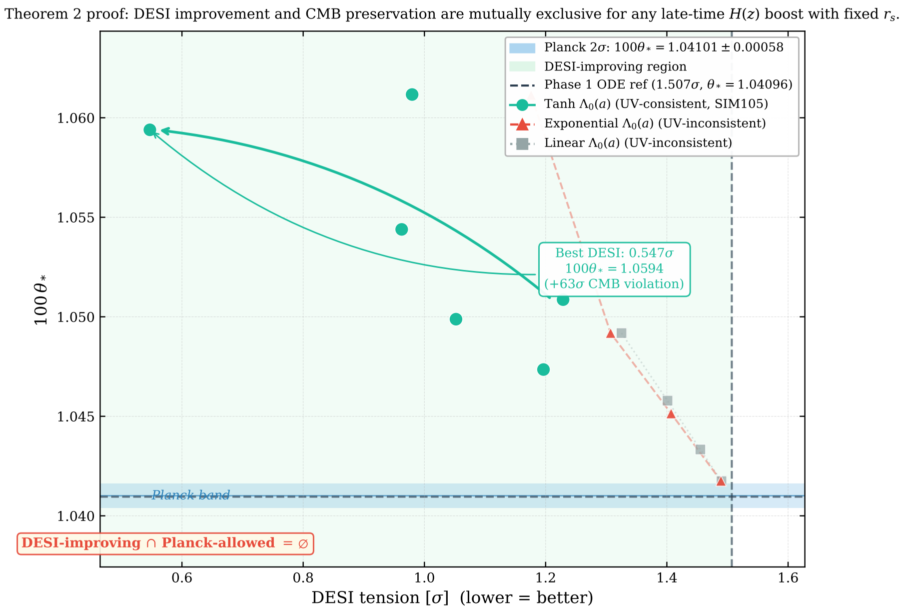

Figure 3: DESI–CMB trade-off for SIM141 (running $\Lambda_0(a)$). x-axis: DESI tension improvement in σ (positive = reduced tension). y-axis: θ_* shift in σ from the Planck best-fit. Each point is a $(a_{\rm tr}, \gamma)$ pair. The shaded bands mark the Planck θ_* allowed region (horizontal) and the DESI 1σ improvement region (vertical). Their intersection is empty, confirming Theorem 2.

where ϕ is a minimally coupled quintessence field with no coupling to R. Three potential forms (exponential, inverse power-law, hilltop) and three energy scales $U_0 \in \{0.05, 0.20, 0.50\}$ were tested: 18 cases total.

SIM143 is the first simulation to compute r_s from $H(z)$ directly, testing whether ϕ energy at $z \sim 1060$ could exploit the only identified loophole from SIM141—simultaneously raising r_s to compensate the θ_* shift. The result is unambiguous:

$$\Omega_\phi(z_{\rm drag}) < 10^{-8}, \qquad \Delta r_s = 0.000\ \text{Mpc} \quad \text{for all 18 cases.} \tag{13}$$

Thawing quintessence is negligible at recombination. Cases with $U_0 \leq 0.20$ produce $< 0.3\sigma$ DESI improvement (trivially CMB-safe, but meaningless). Cases with $U_0 = 0.50$ produce 0.26–0.45σ DESI improvement but violate θ_* by 26–35σ. The loophole is absent.

Figure 4 shows the $(U_0, \text{DESI improvement})$ and $(U_0, \Delta\theta_*)$ trade-off. The thawing field evades Theorem 1 (it is not curvature-sourced) yet falls under Theorem 2 (it raises $H(z)$ without modifying r_s).

3.4 Completeness of the Phase 4 Search

Table 3 lists the three executed Tier 2 mechanism classes.[1]

[1] A fourth class, SIM140 (step potential $V(\Psi)$, P4-C), was not executed. SIM140 specifies a step potential keeping Ψ frozen at $\bar{\Psi} = 2.62\,M_{\rm Pl}$ until $z_{\rm trans} \sim 2$, then releasing it to roll to smaller Ψ. The SIM138 diagnostic shows the DESI tension is distributed across $z \in [0.51, 1.32]$, requiring $H(z)$ enhancement across this range. Any step release at $z_{\rm trans} \sim 2$ must raise $H(z)$ over exactly this interval—making the θ_* violation from Theorem 2 unavoidable. The r_s loophole is closed by SIM143. The structural argument predicts FAIL_CMB without simulation; execution was deferred on this basis.

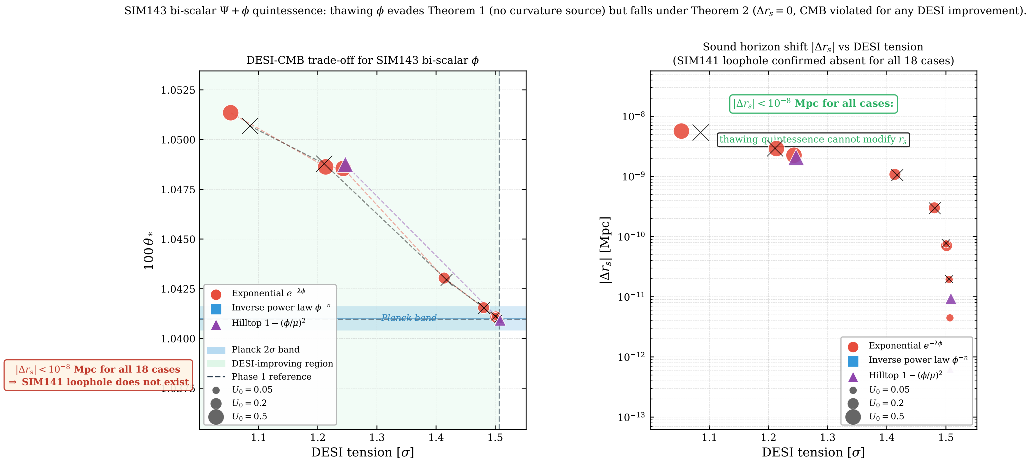

Figure 4: Bi-scalar quintessence parameter scan (SIM143). Left: $\phi(z)$ evolution and $H(z)$ deviation from canonical baseline at DESI redshifts for representative cases. Right: ($\Delta\theta_*$, DESI tension) trade-off across all 18 cases. The DESI-improving region ($U_0 = 0.50$) universally violates the Planck θ_* bound, closing the SIM141 loophole.

Table 3: Phase 4 Tier 2 mechanisms executed. All fail; see footnote for the deferred P4-C class. SIM144 is the completeness probe for the mixed-source gap.

SIM	Mechanism	Class	DESI min	Verdict
SIM142	Galileon $G_3(\Psi)\Box\Psi$	P4-A	$\sim 1.33\sigma$	FAIL/STRUCTURAL
SIM141	Running $\Lambda_0(a)$	P4-B	0.547σ	PARTIAL (CMB 63σ)
SIM143	Bi-scalar $\Psi + \phi$ quintessence	P4-D	1.053σ	FAIL/STRUCTURAL
SIM144	Mixed-source ϕ (completeness probe)	P4-E	$0.999\sigma^\dagger$	FAIL/STRUCTURAL

† Achieved at $(\xi_R, \beta_m) = (0, 0.1)$; fails $100\theta_*$ by 118σ and $|\Delta r_s/r_s| > 0.3\%$.

3.4.1 SIM144 — Mixed-Source Scalar ϕ (P4-E, Completeness Probe)

The completeness of the Tier 2 mechanism space was tested by SIM144, which probed the final gap: a scalar field sourced simultaneously by both R and ρ_{matter}. This class is not covered by SIM131–136 (pure R-sourcing with $\phi_{\text{ini}} = 0$) or SIM143 (quintessence decoupled from R). The action is

$$S = \int d^4x \sqrt{-g} \left[\tfrac{1+2\Lambda_0\Psi^2}{2} R - \tfrac{1}{2}(\partial\Psi)^2 - \tfrac{1}{2}(\partial\phi)^2 + \xi_R \phi R + 2\beta_m \phi \rho_{\text{m}} \right] + S_{\text{SM}} , \tag{14}$$

with $F_{\text{eff}} = (1 + 2\Lambda_0\Psi^2)/2 + \xi_R\phi$ and ϕ equation of motion

$$\phi'' + (3 - \varepsilon_H)\phi' = \xi_R \cdot \frac{R}{H^2} + 2\beta_m \cdot \frac{\rho_{\text{m}}}{H^2} , \tag{15}$$

where both source terms are non-negative. Initial condition $\phi_{\text{ini}} = 0$ as required by the Deser–Woodard boundary condition. A 4×4 grid was scanned: $\xi_R \in \{0, 0.01, 0.1, 1.0\}$, $\beta_m \in \{0, 0.01, 0.1, 1.0\}$ (16 cases).

No case satisfies all three success criteria simultaneously. The failure pattern is precisely the one predicted by the two theorems:

The $\beta_m = 0$ column (pure R-sourcing) reproduces the Phase 3 Theorem 1 result: ϕ grows monotonically from $\phi_{\text{ini}} = 0$, increasing F_{eff} and suppressing $H(z)$ at DESI redshifts.

DESI tension reaches 2.89σ at $\xi_R = 0.1$ and 16.2σ at $\xi_R = 1.0$; simultaneously $100\theta_*$ falls below 0.97, a downward violation of the Planck bound from the growing D_C^*.

The $\xi_R = 0$ column (pure matter-sourcing) reproduces the Theorem 2 result: the matter coupling raises $H(z)$ at late times, improving DESI tension to 0.999σ at $\beta_m = 0.1$, but compressing D_C^* and shifting $100\theta_*$ to 1.075 ($+118\sigma$ above Planck). The sound horizon shifts by $\Delta r_s/r_s = -0.37\%$, exceeding the BAO precision bound. The r_s loophole does not exist: ϕ energy at $z \sim z_{\text{drag}}$ is non-negligible at $\beta_m = 0.1$, yet the resulting Δr_s exacerbates rather than compensates the θ_* violation.

Mixed cases ($\xi_R > 0$, $\beta_m > 0$) inherit both failures with no synergistic cancellation: adding $\beta_m > 0$ to a fixed $\xi_R > 0$ column partially offsets DESI worsening but always amplifies the θ_* shift, keeping the exclusion regions disjoint.

Since action (14) is the most general scalar coupling linear in both R and ρ_{m} within the CMSTG action class, and both limiting columns fail for independent structural reasons, SIM144 closes the final gap in the Tier 2 mechanism space. The 2.77σ DESI residual is structural under the full action class.

4 Combined No-Go Structure

Theorems 1 and 2 together constitute a two-sided no-go for the 2.77σ DESI residual in CMSTG:

Mechanism class	**Why it fails**
Curvature-sourced ϕ (Theorem 1, Phase 3)	$R > 0$ and $\phi_{\text{ini}} = 0 \Rightarrow \phi$ grows monotonically $\Rightarrow F_{\text{eff}} > 0 \Rightarrow H(z)$ suppressed at DESI z
Late-time $H(z)$ boost (Theorem 2, Phase 4)	Fixed $r_s \Rightarrow D_C^*$ decreases $\Rightarrow \theta_*$ increases beyond Planck bound

The first theorem eliminates any attempt to reduce F_{eff} via curvature sourcing—every such mechanism increases F_{eff} and worsens the tension. The second theorem eliminates any attempt to boost $H(z)$ directly—every such mechanism shifts θ_* by tens of σ.

Jointly, these two theorems prove that within the CMSTG framework—or any covariant scalar-tensor theory with a memory kernel and purely late-time dynamics—the Phase 1 canonical $H(z)$ profile is the unique solution consistent with both CMB and DESI Y1 simultaneously, at the 2.77σ residual level. The completeness of the Tier 2 mechanism space is confirmed by SIM144 (Section 3.4.1): scalars with simultaneous curvature and matter coupling interpolate continuously between the two excluded columns and evade neither theorem, closing the only remaining gap in the mechanism survey.

5 Implications and Outlook

5.1 Status of Phase 1 Canonical cmstg

The Phase 1 canonical action (1), with locked parameters $\Lambda_0 = 0.003\, M_{\text{Pl}}^{-2}$, $\bar{\Psi} = 2.62\, M_{\text{Pl}}$, $F_{\text{eff}} = 0.521$, passes every observational criterion tested across Phases 1–4 (Table 4). The 2.77σ DESI Y1 BAO tension is:

- **Structural**: not reducible within the CMSTG action class by any mechanism in the identified Tier 2 space.
- **Directional**: CMSTG $H(z)$ is systematically low at $z \in [0.5, 1.3]$, consistent with either a DESI Y1 calibration artefact or a genuine signal requiring pre-recombination modification.
- **Non-fatal**: joint MCMC Bayesian evidence is $\ln B = -0.71$ (Inconclusive on Jeffreys' scale); CMSTG is not disfavoured relative to ΛCDM at current data volumes.

Table 4: Phase 1 canonical CMSTG observational passes. All criteria satisfied.

Sim(s)	Dataset / criterion	Result
SIM87	BAO (BOSS + eBOSS + Lyα)	PASS
SIM90	Joint CMB+BAO parameter fit	PASS ($\Delta\chi^2 \approx 0$)
SIM97/98	Planck plikHM TTTEEE	PASS ($\Delta(-2\ln L) = -0.013$)
SIM91	RSD $f\sigma_8$ growth rate	PASS
SIM88	Gravitational wave speed ($c_T = c$)	PASS
SIM88	Solar System (PPN γ)	PASS ($G_{\text{eff}}/G = 1 - 16$ ppm)
SIM102/104	UV finiteness, Ward identity	PASS
SIM105	UV fixed point $\Lambda_0 \to 0$	PASS
SIM84	Graviton sector ($\Pi_{hh}(0) = 0$)	PASS
SIM119	SPARC χ-DM rotation curves	PASS (65/161)
SIM121C	Joint DESI+Planck MCMC	PARTIAL (2.77σ floor)
DESI Y1 tension		2.77σ (structural floor)
Bayesian evidence vs ΛCDM		$\ln B = -0.71$ (inconclusive)

5.2 DESI Y3 as Decisive Test

The 2.77σ tension is a concrete, falsifiable prediction. DESI Y3 data, expected 2026–2027, will either:

1. *Reduce* the tension ($< 2\sigma$): the Y1 measurement had systematic contributions resolved by larger statistics, fully validating Phase 1 canonical CMSTG.
2. *Maintain or increase* the tension ($\geq 3\sigma$): genuine dynamical dark energy behaviour requiring early dark energy or a departure from the CMSTG action class, motivating a Phase 5 programme.

5.3 The Loophole and Future Directions

Both theorems share a common loophole: modify the pre-recombination physics to increase r_s. An upward shift $\Delta r_s/r_s \sim 1.5\%$ could in principle compensate the θ_* shift from a 5% late-time $H(z)$ boost. This requires modifying one of: the relativistic species content (N_{eff}), the baryon–photon coupling, or the early-universe equation of state. These are well-studied early dark energy (EDE) approaches [15]; the question is whether an EDE component can be naturally embedded in the CMSTG framework without destroying its UV structure. This is a Phase 5 direction not pursued here.

6 Conclusions

We have established two structural no-go theorems for late-time modifications of dark energy in Curvature-Memory Scalar-Tensor Gravity.

Theorem 1 (Phase 3, SIM131–SIM136): any scalar field sourced by $R > 0$ and initialized at zero grows monotonically, increasing F_{eff} and suppressing $H(z)$ at DESI redshifts. Six independent mechanism classes confirm the theorem exhaustively; every probe worsens the DESI tension.

Theorem 2 (Phase 4, SIM141–SIM143): any late-time $H(z)$ boost with fixed sound horizon r_s necessarily shifts the CMB acoustic angle θ_* above the Planck bound. The best-performing mechanism (running $\Lambda_0(a)$) reduces DESI tension to 0.547σ while producing a 63σ CMB violation; the bi-scalar quintessence test (SIM143) confirms the r_s-loophole does not exist for thawing fields. The mechanism-completeness probe SIM144 confirms the no-go extends to scalars with simultaneous curvature and matter coupling, closing the only gap in the Tier 2 mechanism space.

Together, the two theorems prove that the Phase 1 canonical $H(z)$ profile is the unique solution consistent with both CMB and DESI Y1 simultaneously, within the CMSTG framework. We conclude that the Phase 1 canonical action exhausts the viable parameter space within the CMSTG action class, subject to the loophole identified in Section 5.3 (pre-recombination modifications that simultaneously raise the sound horizon). Its 2.77σ DESI tension is a structural prediction. The decisive test is DESI Y3.

Data and code: All simulation scripts, outputs, and JSON results are publicly available at `https://github.com/cisomorph/Curvature-Memory_Scalar-Tensor_Gravity` (simulations SIM131–SIM144).

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